

SQL Assignment: Farmers Insurance Analysis

Team :

1. Anusha T
2. Maaz Hussain
3. Sri Lekha
4. Suraj Singh Chauhan

GITHUB LINK:- https://github.com/surajra123/SQL_Farmers_Insurance_Analysis

This project analyses the dataset using SQL. The dataset was imported into MySQL, and data cleaning was performed to handle NULL values and ensure consistency.

I. SELECT Queries

- Retrieved all 27 states.
- Extracted total farmers covered and sum insured per state

srcStateName varchar	TotalFarmersCovered decimal	SumInsured double
MADHYA PRADESH	9149678	8413198.288650513
MAHARASHTRA	8939015	4005734.66847685
UTTAR PRADESH	7239576	3298970.927065717
RAJASTHAN	6822958	5363500.742112458
TAMIL NADU	3962760	3025519.4509361982
ODISHA	2586332	1375248.5827995362
HARYANA	2081990	2268958.5814785063
WEST BENGAL	2054414	842893.1048812866
CHHATTISGARH	1858973	1118190.1713431925
ANDHRA PRADESH	1810317	1631182.105102539

II. Filtering Data

- Extracted records for year 2020
- Filtered to 4 records for state Himachal Pradesh with TotalPopulationRural > 10,00,000
- Calculated district-wise premium sums

srcStateName varchar	TotalFarmersCovered decimal	TotalGrossPremiumA double
JAMMU AND KASHMIR	52842	2472.040069580078
HIMACHAL PRADESH	64766	530.0900039672852
UTTARAKHAND	19726	668.5700073242188
HARYANA	665511	26875.559646606445
RAJASTHAN	1666703	114021.20956420898
TELANGANA	150069	4464.029975891113
UTTAR PRADESH	2614548	70172.93009567261
ASSAM	38635	607.4599914550781
WEST BENGAL	2038418	49236.04949951172
JHARKHAND	25538	462.1999969482422
ODISHA	58010	833.5800018310547
CHHATTISGARH	151742	6863.970027923584
MADHYA PRADESH	2523882	155228.18968582153
MAHARASHTRA	2425998	90419.19887542725
ANDHRA PRADESH	473849	36850.88082885742
KERALA	5044	270.67999267578125
TAMIL NADU	1134011	157822.25834655762

III. Aggregation (GROUP BY)

- Calculated average insured land area per year

srcYear int	AverageInsuredLand double
2020	46.92393946313323
2018	37.878552367563046
2019	38.264534640478175
2021	38.54617980123473

- Computed total farmers covered per district

- Identified states with high total insured sums

srcStateName varchar	TotalFarmersPremium double	TotalStatePremiumA double	TotalGOVPremiumAn double	TotalFarmersCovered decimal
HARYANA	49235.0031671694	60606.83240488474	60596.915601226734	2081990
RAJASTHAN	104352.47155295964	290076.45928888395	276115.27905206755	6822958
UTTAR PRADESH	60333.822952648974	88797.06063364266	88694.36060617684	7239576
ASSAM	850.0539912581444	19693.19951307727	13291.469552039634	1590716
WEST BENGAL	8727.983597874641	25480.18007349968	15393.359973311424	2054414
ODISHA	26987.440245104488	112098.36936168672	112098.36936168672	2586332
CHHATTISGARH	20947.702994682593	68043.93652160466	68043.93652160466	1858973
MADHYA PRADESH	139861.34987795353	399901.49423742294	399901.49423742294	9149678
MAHARASHTRA	84293.27857522154	376644.25065794867	356388.9691027729	8939015
ANDHRA PRADESH	5555.080860212445	94441.73965454102	57365.21954727173	1810317
TAMIL NADU	51184.67832771689	427455.7829892859	306309.6836909279	3962760

IV. Sorting Data (ORDER BY)

- Identified top 5 districts by population in 2020.

srcDistrictName varchar	TotalPopulation int
Bengaluru Urban	9621551
Pune	9429408
Thane	8070032
Jaipur	6626178
Nashik	6107187

- Found districts with lowest premium amounts

srcStateName varchar	srcDistrictName varchar	SumInsured float
KARNATAKA	Kalaburgi	0.0044
KARNATAKA	Kolar	0.0048
KARNATAKA	Davangere	0.0051
KARNATAKA	Ballari	0.0052
KARNATAKA	Mandya	0.0056
KARNATAKA	YADGIRI	0.0089
KARNATAKA	Bidar	0.0103
KARNATAKA	Ballari	0.0107
KARNATAKA	Davangere	0.0122
KARNATAKA	Koppal	0.0135

- Ranked states based on farmer-to-population ratio

srcStateName varchar	srcYear int	InsuredFarmersRatio decimal
CHHATTISGARH	2021	0.0497
TRIPURA	2020	0.0468
TRIPURA	2021	0.0464

V. String Functions

- Extracted short state names

srcStateName varchar	StateShortName varchar
KARNATAKA	KAR
JAMMU AND KASHMIR	JAM
HIMACHAL PRADESH	HIM
UTTARAKHAND	UTT
HARYANA	HAR
GUJARAT	GUJ

- Filtered 60 districts starts with "B" and 59 districts ends with "pur".

VI. Joins

- Combined datasets to analyse premium distribution.
- Identified 34 districts with highest premium values.
- Compared population and premium data

srcStateName varchar	srcDistrictName varchar	TotalPremiumAmoun double	AvgPopulation decimal
MAHARASHTRA	Bid	14528.859741210938	2583421.0000
MADHYA PRADESH	Ujjain	12385.230346679688	1986864.0000
MAHARASHTRA	Latur	11700.31982421875	2454196.0000

VII. Subqueries

- Compared district performance against averages
- Identified high-performing states and districts(166)

VIII. Window Functions

- Ranked records using ROW_NUMBER and RANK
- Calculated cumulative premiums

IX. Data Integrity

X. UPDATE and DELETE

Tables were created with primary and foreign keys to maintain data consistency. Update and delete operations were performed to modify and clean the dataset.